

Building a Chatbot in Angular

Using Dialogflow



A chatbot simulates conversations with human users, especially over the Internet. This can be in the form of voice or text messages. Chatbots are being increasingly used in commercial environments. Dialogflow can be used to create a great chatbot. In this tutorial, the author uses the JavaScript SDK to build a chatbot directly in Angular.

Dialogflow, formerly known as API.AI, is an engine for building conversational experiences. It harnesses Google's machine learning and natural language processing technologies to give Web developers a user friendly interface to build chatbots and deploy them on any platform, including the Web, on Slack or Facebook Messenger.

Prerequisites

Let's start by building a new Angular app from the command line using `ng new chatbot`, and then 'cd' into that folder to install the JavaScript SDK via NPM in the development environment.

```
npm install -g @angular/cli
ng new chatbot
cd chatbot
npm install api-ai-javascript --save-dev
```

Note: At this point, the SDK is still called API.AI, but it may change to Dialogflow in the future.

I am using Visual Studio code for this project. You are free to choose your own code editor.

Feature module (*chat.module.ts*)

Technically, this step is optional, but it is good practice and will make your life a lot easier as your app grows more complex. The first step is to generate a module called *chat*, as follows:

```
ng g module chat
```

Then you can add resources to this module, like a service, and you can use the `-m` flag to make sure that it's included in this module.

```
ng g service chat -m chat
```

In addition, you need to create a component called *chat-dialog* and place that in the module as well, as shown below:

```
ng g component chat/chat-dialog -m chat
```

You can see in the file structure that all your resources are nested under the *chat* folder, and the components and services are registered inside the *chat* module. The only extra bit that you need to add is the Angular *FormsModule* and set your *ChatDialogComponent* to *Exports* so that it can be used directly in the app component. The *chat.module.ts* file should look like what's shown below:

```
import { NgModule } from '@angular/core';
import { CommonModule } from '@angular/common';
import { FormsModule } from '@angular/forms';
import { ChatService } from '../chat.service';
import { ChatDialogComponent } from '../chat-dialog/chat-dialog.component';
```

```
@NgModule({
  imports: [
    CommonModule,
    FormsModule
```